Midterm Exam

FinTech 545 — Quantitative Risk Management

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Instructions

Be verbose. Explain clearly your reasoning, methods, and results in your written work. No code is necessary in the write up, but including it could result in partial credit.

Written answers are worth the amounts stated. **Total points available:** 120. The exam is graded out of 100. Question 3 is long, and 20 of its points are extra credit.

General Guidelines

1. **Format:** Answers should be formatted as a PDF. You may convert your Python notebook, if you use one, directly to PDF. Alternatively, use an editor like Word and print to PDF.
2. **Restate the question:** Include the question number and restate each question before your answer.
3. **Submission:** When finished, email your PDF, your code, and the PDF of your AI prompts (if you used AI) to the TAs and to me.
4. **Label your code:** Clearly mark which problem and part each section of your code answers, for example with a comment such as `# Problem 2(a)`. We need to be able to find the code behind every answer.
5. **Repository:** Do not check code or answers into your repository until after the exam is completed by all.
6. **Data:** Data for each problem is in the repository, named by the question number.
7. **Conventions:** Use arithmetic returns. Report VaR and ES as positive numbers for a loss.
8. **Permitted resources:**
    - You may use your notes and the internet.
    - You may use generative AI to help write your code. You may not use it to answer the written questions. Do not ask an AI to explain a result, choose a model, or give you the reasoning for an answer. The reasoning must be your own.
    - Tell your AI agent at the start of the session that it may only write code.
    - Keep a separate document with every prompt you gave the AI and every response it returned. You do not need to include the code it wrote. Convert this document to a PDF and include it in your submission email.
    - You are responsible for everything in your code, including anything an AI wrote for you.
    - You may not work with other students. All work must be your own.
9. **Honor Code:** All students will be held to the Duke Community Standard.

Duke Community Standard

Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Citizens of this community commit to reflect upon and uphold these principles in all academic and nonacademic endeavors, and to protect and promote a culture of integrity.

To uphold the Duke Community Standard:

- I will not lie, cheat, or steal in my academic endeavors;
- I will conduct myself honorably in all my endeavors; and
- I will act if the Standard is compromised.

1 Pairwise Correlation with Missing Data (25 pts)

Using `problem1.csv`:

The file holds 50 days of returns for five series, `x1` through `x5`. A blank cell is a missing value. Calculate the correlations pairwise, and use the full sample of each series to calculate its variance.

a. (5) Calculate the pairwise correlation matrix.
b. (3) Calculate the variance of each series, using every observation that series has.
c. (5) Is the correlation matrix positive semi-definite? Show how you know.
d. (7) If it is not, use Higham's method to find the nearest positive semi-definite correlation matrix. Report the matrix and its eigenvalues. Which entries changed the most?
e. (5) Combine your results into a covariance matrix. Report it.

2 VaR and ES for a Single Stock (20 pts)

Using `problem2.csv`:

The file holds 501 daily prices for the stock `XYZ` and for `SPY`. You hold 100 shares of `XYZ`.

You believe that `XYZ` is related to the market (`SPY`) by the following equation:

$$r_{XYZ} = \alpha + \beta\, r_{SPY} + \epsilon$$

You assume the arithmetic returns of `SPY` and ϵ are normally distributed.

a. (10) What is your VaR at the 95% and 99% levels, in percent? Explain your method.
b. (10) What is your ES at the 95% and 99% levels, in percent? Explain your method.

3 Copula VaR and ES for a Two Book Portfolio (75 pts, includes 20 extra credit)

Using `problem3.csv` and `problem3_holdings.csv`:

`problem3.csv` holds 1,001 daily prices for ten stocks, `S1` through `S10`. `problem3_holdings.csv` gives the number of shares you hold of each stock and the book it belongs to. The stocks are split between two books, `A` and `B`. Value each position at the last price in the file.

a. (25) For each stock, fit a normal, a Student's t, and a normal inverse Gaussian distribution to its returns. Report the fitted parameters. Which distribution fits each stock best? Justify your choice.
b. (20) Using the distributions you chose in part (a), fit a Gaussian copula and a Student's t copula. Which one fits better? Justify your choice.
c. (30) Using the copula you chose in part (b), simulate the portfolio with at least 5,000 draws. Calculate the absolute VaR and ES at the 95% level, in dollars, for each stock, each book, and the total portfolio. Present the results in a table. Discuss the results.